

1) Docker Image

1.1

```
juan@192.168.20.22's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

336 updates can be applied immediately.
267 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

47 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Apr 17 13:51:43 2026 from 192.168.20.5
juan@juan-virtualbox:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	7f0adca1fc6c	240MB	65.8MB	

```
juan@juan-virtualbox:~$
```

1.2

```
Last login: Fri Apr 17 13:51:43 2026 from 192.168.20.5
juan@juan-virtualbox:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	7f0adca1fc6c	240MB	65.8MB	

```
juan@juan-virtualbox:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
juan@juan-virtualbox:~$ cd
juan@juan-virtualbox:~$ cd Desktop/
juan@juan-virtualbox:~/Desktop$ mkdir docker-lab
juan@juan-virtualbox:~/Desktop$ cd docker-lab/
juan@juan-virtualbox:~/Desktop/docker-lab$ nano dockerfile
juan@juan-virtualbox:~/Desktop/docker-lab$ docker build -t miapp:v1 .
[+] Building 65.3s (7/7) FINISHED
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 147B
=> [internal] load metadata for docker.io/library/alpine:3.20
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/3] FROM docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
=> => resolve docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
=> => sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd9f98b50d2e1b96130471 3.63MB / 3.63MB
=> => extracting sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd9f98b50d2e1b96130471
=> [2/3] WORKDIR /app
=> [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:2e7f9bbe390ebe11cdae1ab7a6a9dfaa269642d1c9b736a06ded5f4d874d6b35
=> => exporting config sha256:e291ba6056d6d5f8e24b0887891e428033a11cc594076b61aed00df9ed30ddc9
=> => exporting attestation manifest sha256:07cd214e1ad683e66e19b8a4abfaae78158761a825710b293aab1e895d042584
=> => exporting manifest list sha256:9a3eb4f83bf953c7d83d57707f5a51ee83a3dc4fe6b4bd8c2caf1b3da636a3f6
=> => naming to docker.io/library/miapp:v1
=> => unpacking to docker.io/library/miapp:v1
juan@juan-virtualbox:~/Desktop/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1          9a3eb4f83bf9      12.1MB      3.63MB
juan@juan-virtualbox:~/Desktop/docker-lab$ docker run --rm miapp:v1
Hola desde Docker
juan@juan-virtualbox:~/Desktop/docker-lab$
```

1.3

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker tag miapp:v1 miapp:latest
juan@juan-virtualbox:~/Desktop/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest    9a3eb4f83bf9    12.1MB    3.63MB
miapp:v1        9a3eb4f83bf9    12.1MB    3.63MB
juan@juan-virtualbox:~/Desktop/docker-lab$
```

1.4

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker image inspect miapp:v1
[
  {
    "Id": "sha256:9a3eb4f83bf953c7d83d57707f5a51ee83a3dc4fe6b4bd8c2caf1b3da636a3f6",
    "RepoTags": [
      "miapp:latest",
      "miapp:v1"
    ],
    "RepoDigests": [
      "miapp@sha256:9a3eb4f83bf953c7d83d57707f5a51ee83a3dc4fe6b4bd8c2caf1b3da636a3f6"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-04-17T15:22:51.476732888-05:00",
    "Config": {
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
      ],
      "Cmd": [
        "cat",
        "/app/mensaje.txt"
      ],
      "WorkingDir": "/app",
      "ArgsEscaped": true
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 3633527,
    "RootFS": {
      "Type": "layers",
      "Layers": [
        "sha256:08bc4e534116aa76b16015484b82eac51f9a593416feae9296c8a2d4bb7aa4a2",
        "sha256:36f3bb68c96ccd5b0e8c2e7acde69e8fbc9a6fb397d5e991011b62bad656367c",
        "sha256:2f30c73720573c09533155ec533e5b29942fd7ec6e37daa0cd5e9bdbd4b0f27e"
      ]
    },
    "Metadata": {
      "LastTagTime": "2026-04-17T20:24:58.337669858Z"
    },
    "Descriptor": {
      "mediaType": "application/vnd.oci.image.index.v1+json",
      "digest": "sha256:9a3eb4f83bf953c7d83d57707f5a51ee83a3dc4fe6b4bd8c2caf1b3da636a3f6",
      "size": 855
    },
    "Identity": {
      "Build": [
        {
          "Ref": "jsj1fdy4tgfizzj92idaj3xwn",
          "CreatedAt": "2026-04-17T15:22:52.154785807-05:00"
        }
      ]
    }
  }
]
juan@juan-virtualbox:~/Desktop/docker-lab$ docker image inspect miapp:v1 --format '{{.Id}}'
sha256:9a3eb4f83bf953c7d83d57707f5a51ee83a3dc4fe6b4bd8c2caf1b3da636a3f6
juan@juan-virtualbox:~/Desktop/docker-lab$
```

1.5

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker save -o miapp.tar miapp:v1
juan@juan-virtualbox:~/Desktop/docker-lab$ ls -lh miapp.tar
-rw----- 1 juan juan 3,5M abr 17 15:26 miapp.tar
juan@juan-virtualbox:~/Desktop/docker-lab$ |
```

1.6

```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker rmi miapp:v1
Untagged: miapp:v1
juan@juan-virtualbox:~/Desktop/docker-lab$ docker load -i miapp.tar
docker: command not found
juan@juan-virtualbox:~/Desktop/docker-lab$ docker load -i miapp.tar
Loaded image: miapp:v1
juan@juan-virtualbox:~/Desktop/docker-lab$ docker images | grep miapp
docker: command not found
juan@juan-virtualbox:~/Desktop/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest  9a3eb4f83bf9  12.1MB  3.63MB
miapp:v1      9a3eb4f83bf9  12.1MB  3.63MB
juan@juan-virtualbox:~/Desktop/docker-lab$

```

1.7

```

miapp:v1 9a3eb4f83bf9 12.1MB 3.63MB
juan@juan-virtualbox:~/Desktop/docker-lab$ docker rmi miapp:latest
Untagged: miapp:latest
juan@juan-virtualbox:~/Desktop/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1 9a3eb4f83bf9 12.1MB 3.63MB
juan@juan-virtualbox:~/Desktop/docker-lab$

```

1.8

```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker image prune -f
Total reclaimed space: 0B
juan@juan-virtualbox:~/Desktop/docker-lab$ |

```

2) Docker Container

2.1

```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker run -d --name web1 nginx
470bc894aa02b5bb4836e4412aa311aab8346d4fe998761d444737d7b4bd6600
juan@juan-virtualbox:~/Desktop/docker-lab$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS      PORTS      NAMES
470bc894aa02   nginx    "/docker-entrypoint..." 8 seconds ago  Up 7 seconds  80/tcp     web1

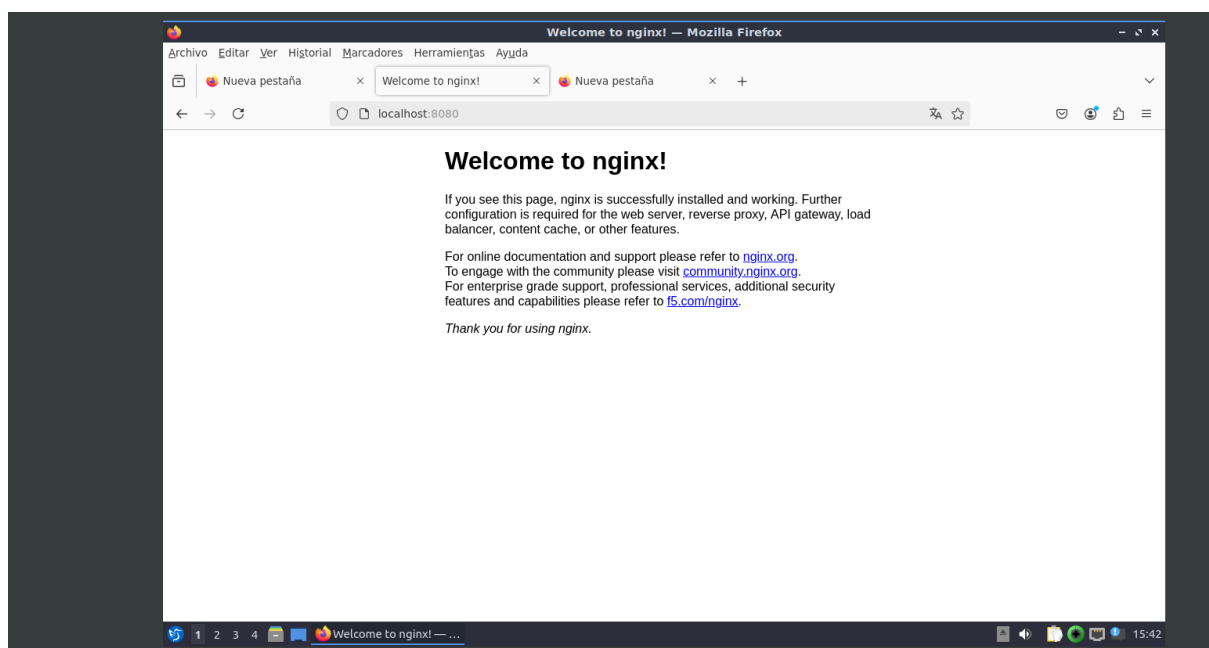
```

2.2

```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker run -d --name web2 -p 8080:80 nginx:latest
574353ab63f38266a445b6aab081f33f49c2108fd656655480cf5955679e6715
juan@juan-virtualbox:~/Desktop/docker-lab$ |

```



2.3

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
574353ab63f3   nginx:latest "/docker-entrypoint..." 7 minutes ago Up 7 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
470bc894aa02   nginx     "/docker-entrypoint..." 12 minutes ago Up 12 minutes   80/tcp                               web1
juan@juan-virtualbox:~/Desktop/docker-lab$
```

2.4 y 2.5

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker logs web2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/17 20:36:21 [notice] 1#1: using the "epoll" event method
2026/04/17 20:36:21 [notice] 1#1: nginx/1.29.8
2026/04/17 20:36:21 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/04/17 20:36:21 [notice] 1#1: OS: Linux 6.8.0-40-generic
2026/04/17 20:36:21 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/04/17 20:36:21 [notice] 1#1: start worker processes
2026/04/17 20:36:21 [notice] 1#1: start worker process 29
172.17.0.1 - - [17/Apr/2026:20:37:49 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (X11; Ubuntu
2026/04/17 20:37:50 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No suc
:8080", referer: "http://localhost:8080/"
172.17.0.1 - - [17/Apr/2026:20:37:50 +0000] "GET /favicon.ico HTTP/1.1" 404 153 "http://localhost:
juan@juan-virtualbox:~/Desktop/docker-lab$ docker inspect web2 --format '{{.State.Status}}'
running
juan@juan-virtualbox:~/Desktop/docker-lab$
```

2.6

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker stop web2
web2
juan@juan-virtualbox:~/Desktop/docker-lab$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
juan@juan-virtualbox:~/Desktop/docker-lab$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
574353ab63f3   nginx:latest "/docker-entrypoint..." 10 minutes ago Exited (0) 13 seconds ago   web2
470bc894aa02   nginx     "/docker-entrypoint..." 15 minutes ago Exited (0) 2 minutes ago   web1
juan@juan-virtualbox:~/Desktop/docker-lab$
```

```
470bc894aa02   nginx     "/docker-entrypoint..." 15 minutes ago Exited (0) 2 minutes ago   web1
juan@juan-virtualbox:~/Desktop/docker-lab$ docker start web2
web2
juan@juan-virtualbox:~/Desktop/docker-lab$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
574353ab63f3   nginx:latest "/docker-entrypoint..." 12 minutes ago Up 3 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
470bc894aa02   nginx     "/docker-entrypoint..." 17 minutes ago Exited (0) 3 minutes ago   web1
juan@juan-virtualbox:~/Desktop/docker-lab$
```

2.7

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
Digest: sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
Status: Downloaded newer image for alpine:3.20
/ # echo "prueba interactiva"
prueba interactiva
/ # exit
juan@juan-virtualbox:~/Desktop/docker-lab$
```

2.8 y 2.9

```
470bc894aa02   nginx     "/docker-entrypoint..." 31 minutes ago Exited (0) 18 minutes ago   web1
juan@juan-virtualbox:~$ docker rm alpine1
alpine1
juan@juan-virtualbox:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
574353ab63f3   nginx:latest "/docker-entrypoint..." 27 minutes ago Exited (0) 4 minutes ago   web2
470bc894aa02   nginx     "/docker-entrypoint..." 32 minutes ago Exited (0) 18 minutes ago   web1
juan@juan-virtualbox:~$ docker start web2
web2
juan@juan-virtualbox:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
574353ab63f3   nginx:latest "/docker-entrypoint..." 27 minutes ago Up 2 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
470bc894aa02   nginx     "/docker-entrypoint..." 32 minutes ago Exited (0) 18 minutes ago   web1
juan@juan-virtualbox:~$ docker rm -f web2
web2
juan@juan-virtualbox:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
470bc894aa02   nginx     "/docker-entrypoint..." 32 minutes ago Exited (0) 19 minutes ago   web1
juan@juan-virtualbox:~$
```

3) Volúmenes

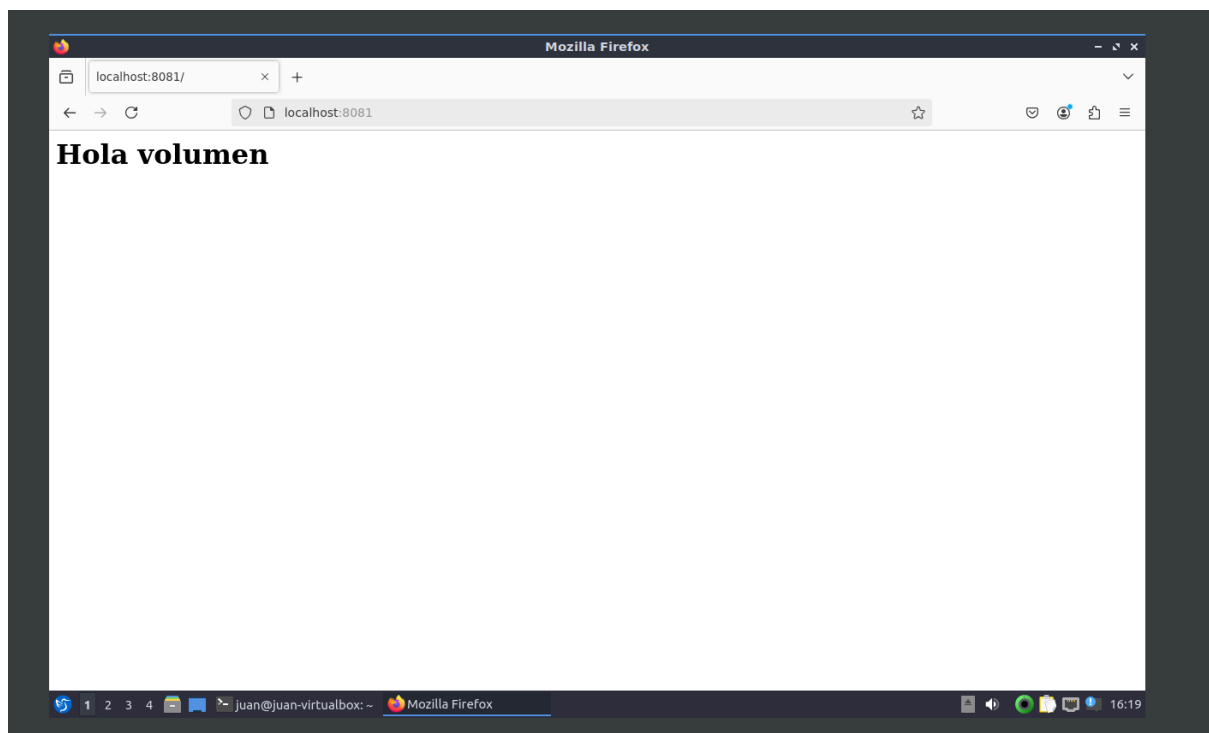
3.1

```
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS
juan@juan-virtualbox:~$ docker volume create datos_nginx
datos_nginx
juan@juan-virtualbox:~$ docker volume ls
DRIVER      VOLUME NAME
local      datos_nginx
juan@juan-virtualbox:~$
```

3.2

```
local      datos_nginx
juan@juan-virtualbox:~$ docker volume inspect datos_nginx
[
  {
    "CreatedAt": "2026-04-17T16:11:33-05:00",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/datos_nginx/_data",
    "Name": "datos_nginx",
    "Options": null,
    "Scope": "local"
  }
]
```

3.3



```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker run -d --name webvol -p 8081:80 -v
flag needs an argument: 'v' in -v

Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Run 'docker run --help' for more information
juan@juan-virtualbox:~/Desktop/docker-lab$ docker run -d --name webvol -p 8081:80 -v datos_nginx:/usr/share/nginx/html nginx:latest
e849162d802d5e51486e0c8d262ccdddba2659ac20db6f22826d958731b18d1c
juan@juan-virtualbox:~/Desktop/docker-lab$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
e849162d802d   nginx:latest   "/docker-entrypoint..." 45 seconds ago Up 45 seconds 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp webvol
juan@juan-virtualbox:~/Desktop/docker-lab$ echo "<h1>Hola volumen</h1>" > index.html
dockerfile  index.html  miapp.tar
juan@juan-virtualbox:~/Desktop/docker-lab$ docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 22B (transferred 2.05kB) to webvol:/usr/share/nginx/html/index.html
juan@juan-virtualbox:~/Desktop/docker-lab$ curl http://localhost:8081
<h1>Hola volumen</h1>

```

3.4

```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker rm -f webvol
webvol
juan@juan-virtualbox:~/Desktop/docker-lab$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
juan@juan-virtualbox:~/Desktop/docker-lab$ docker volume rm datos_nginx
datos_nginx
juan@juan-virtualbox:~/Desktop/docker-lab$ docker volume ls
DRIVER          VOLUME NAME
juan@juan-virtualbox:~/Desktop/docker-lab$ |

```

4) Redes

4.1 y 4.2

```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker network create mi_red
7356b0bc9ebca6714d274951a7b7aea4dff70441db146b93fcf2382019ad5b58
juan@juan-virtualbox:~/Desktop/docker-lab$ docker network ls
NETWORK ID      NAME      DRIVER      SCOPE
e255466563a7    bridge    bridge      local
450622f9a685    host     host        local
7356b0bc9ebc    mi_red    bridge      local
ab6b3ae2ecad    none     null        local
juan@juan-virtualbox:~/Desktop/docker-lab$ |

```

4.3


```

ab0b3ae2ecad none null local
juan@juan-virtualbox:~/Desktop/docker-lab$ docker network inspect mi_red
[
  {
    "Name": "mi_red",
    "Id": "7356b0bc9ebca6714d274951a7b7aea4dfff70441db146b93fcf2382019ad5b58",
    "Created": "2026-04-17T16:24:57.003729342-05:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {},
    "Status": {
      "IPAM": {
        "Subnets": {
          "172.18.0.0/16": {
            "IPsInUse": 3,
            "DynamicIPsAvailable": 65533
          }
        }
      }
    }
  }
]
juan@juan-virtualbox:~/Desktop/docker-lab$

```

4.4

```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker run -d --name caja1 --network mi_red alpine:3.20 sleep 300
f381b090a54782c4746e60f965bf290f757acf4b8f990c16b8051ca369b57355
juan@juan-virtualbox:~/Desktop/docker-lab$ docker run -d --name caja2 --network mi_red alpine:3.20 sleep 300
3f80bf8bfb0fe6570affb5aaf64d1581f2ceaf16bd6c740cd0197050146e721d
juan@juan-virtualbox:~/Desktop/docker-lab$ docker exec caja1 ping -c 2 caja2
PING caja2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.157 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.083 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.083/0.120/0.157 ms
juan@juan-virtualbox:~/Desktop/docker-lab$

```

4.5

```

round-trip min/avg/max = 0.083/0.120/0.157 ms
juan@juan-virtualbox:~/Desktop/docker-lab$ docker network disconnect mi_red caja2
juan@juan-virtualbox:~/Desktop/docker-lab$ docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'
juan@juan-virtualbox:~/Desktop/docker-lab$

```

5) Docker Compose

5.1

```
juan@juan-virtualbox:~/Desktop/docker-lab$ cat compose.yml
services:
  web:
    image: nginx:latest
    ports:
      - "8082:80"
```

5.2

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker compose up -d
[+] up 2/2
  ✓ Network docker-lab_default Created
  ✓ Container docker-lab-web-1 Started
juan@juan-virtualbox:~/Desktop/docker-lab$ docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED      STATUS      PORTS
docker-lab-web-1    nginx:latest      "/docker-entrypoint..." web         9 seconds ago Up 9 seconds 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
juan@juan-virtualbox:~/Desktop/docker-lab$ curl http://localhost:8082
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
juan@juan-virtualbox:~/Desktop/docker-lab$
```

5.3

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker compose logs
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/04/17 21:46:43 [notice] 1#1: using the "epoll" event method
web-1 | 2026/04/17 21:46:43 [notice] 1#1: nginx/1.29.8
web-1 | 2026/04/17 21:46:43 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1 | 2026/04/17 21:46:43 [notice] 1#1: OS: Linux 6.8.0-40-generic
web-1 | 2026/04/17 21:46:43 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
web-1 | 2026/04/17 21:46:43 [notice] 1#1: start worker processes
web-1 | 2026/04/17 21:46:43 [notice] 1#1: start worker process 29
web-1 | 172.19.0.1 - - [17/Apr/2026:21:47:01 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/7.81.0" "-"
juan@juan-virtualbox:~/Desktop/docker-lab$
```

5.4

```
juan@juan-virtualbox:~/Desktop/docker-lab$ docker compose down
[+] down 2/2
  ✓ Container docker-lab-web-1 Removed
  ✓ Network docker-lab_default Removed
juan@juan-virtualbox:~/Desktop/docker-lab$ docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED      STATUS      PORTS
juan@juan-virtualbox:~/Desktop/docker-lab$
```

5.5


```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker compose up -d --scale web=3
[+] up 3/4
  ✓ Network docker-lab_default Created
  ✓ Container docker-lab-web-3 Created
  ✓ Container docker-lab-web-1 Started
  " Container docker-lab-web-2 Starting
Error response from daemon: failed to set up container networking: driver fail
ind for 0.0.0.0:8082 failed: port is already allocated
juan@juan-virtualbox:~/Desktop/docker-lab$ |

```

```

juan@juan-virtualbox:~/Desktop/docker-lab$ docker compose up -d --scale web=3
[+] up 3/4
  ✓ Network docker-lab_default Created
  ✓ Container docker-lab-web-3 Created
  ✓ Container docker-lab-web-1 Started
  " Container docker-lab-web-2 Starting
Error response from daemon: failed to set up container networking: driver faile
ind for 0.0.0.0:8082 failed: port is already allocated
juan@juan-virtualbox:~/Desktop/docker-lab$ docker compose down
[+] down 4/4
  ✓ Container docker-lab-web-3 Removed
  ✓ Container docker-lab-web-1 Removed
  ✓ Container docker-lab-web-2 Removed
  ✓ Network docker-lab_default Removed
juan@juan-virtualbox:~/Desktop/docker-lab$ docker compose ls
NAME                STATUS             CONFIG FILES
juan@juan-virtualbox:~/Desktop/docker-lab$ |

```

6) Dockerfile: probar instrucciones principales

```

juan@juan-virtualbox:~/Desktop/docker-lab$ mkdir dockerfile-test
juan@juan-virtualbox:~/Desktop/docker-lab$ cd dockerfile-test/
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ echo "contenido local" > archivo.txt
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ nano Dockerfile
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ ls
archivo.txt  Dockerfile
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ nano Dockerfile
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[+] Building 0.2s (1/1) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 344B
Dockerfile:11
-----
  9 |     VOLUME ["/data"]
 10 |     CMD ["sh", "-c", "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat
 11 | >>> /app/version.txt"]
 12 |
-----
ERROR: failed to build: failed to solve: dockerfile parse error on line 11: unknown instruction: /app/version.txt"]
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ docker run --rm dockerfile-demo
Unable to find image 'dockerfile-demo:latest' locally
docker: Error response from daemon: pull access denied for dockerfile-demo, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ cat <<EOF > Dockerfile
FROM alpine:3.20
LABEL autor="prueba"
ARG VERSION=1.0
ENV APP_ENV=dev
WORKDIR /app
COPY archivo.txt /app/archivo.txt
RUN echo "version=${VERSION}" > /app/version.txt
ADD archivo.txt /app/archivo2.txt
VOLUME ["/data"]
CMD ["sh", "-c", "echo APP_ENV=${APP_ENV} && cat /app/archivo.txt && cat /app/version.txt"]
EOF
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[+] Building 2.2s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 344B
=> [internal] load metadata for docker.io/library/alpine:3.20
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
=> => resolve docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
=> [internal] load build context
=> => transferring context: 54B
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY archivo.txt /app/archivo.txt
=> [4/5] RUN echo "version=2.0" > /app/version.txt
=> [5/5] ADD archivo.txt /app/archivo2.txt
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:01b6e36c19de446f6caadb7423b3cd0d8dafa838dc77f6fed3e4cf08a25b61bf
=> => exporting config sha256:695d6d64abb5db2a3ecbaa9142018aec0313a3ac51e23613d2e0f13bd9039d6f
=> => exporting attestation manifest sha256:30e715d2407a48e692c41b4f67ebb3db49b6aa779ddc7edb247652cfb5ae3d49
=> => exporting manifest list sha256:61ca7b6974f9e48f045606b4c18f8dec4008789c8da60f6a58de148b1dc72b37
=> => pushing to docker.io/library/dockerfile-demo:latest
=> => naming to docker.io/library/dockerfile-demo:latest
=> => unpacking to docker.io/library/dockerfile-demo:latest
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ docker run --rm dockerfile-demo
APP_ENV=dev
contenido local
version=2.0
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$

```

```

juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ docker image inspect dockerfile-demo
[
  {
    "Id": "sha256:61ca7b6974f9e48f045606b4c18f8dec4008789c8da60f6a58de148b1dc72b37",
    "RepoTags": [
      "dockerfile-demo:latest"
    ],
    "RepoDigests": [
      "dockerfile-demo@sha256:61ca7b6974f9e48f045606b4c18f8dec4008789c8da60f6a58de148b1dc72b37"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-04-17T17:03:04.604423215-05:00",
    "Config": {
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "APP_ENV=dev"
      ],
      "Cmd": [
        "sh",
        "-c",
        "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"
      ],
      "Volumes": {
        "/data": {}
      },
      "WorkingDir": "/app",
      "Labels": {
        "autor": "prueba"
      },
      "ArgsEscaped": true
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 3635425,
    "RootFS": {
      "Type": "layers",
      "Layers": [
        "sha256:08bc4e534116aa76b16015484b82eac51f9a593416fae9296c8a2d4bb7aa4a2",
        "sha256:36f3bb68c96ccd5b0e8c2e7acde69e8fbc9a6fb397d5e991011b62bad656367c",
        "sha256:aab80bab028d3ae724baba7335f22d33d555397b84af493691eae831d4489e6",
        "sha256:5500c6407c8137c0a9643c4a47754ff07d9fe34086b17a39f9ecad61ce8c0eba",
        "sha256:c91d43a942a3776be6cec8b2bc047e1b8b540bfcbcc91ab3265109d76dd068e1"
      ]
    },
    "Metadata": {
      "LastTagTime": "2026-04-17T22:03:05.237677561z"
    },
    "Descriptor": {
      "mediaType": "application/vnd.oci.image.index.v1+json",
      "digest": "sha256:61ca7b6974f9e48f045606b4c18f8dec4008789c8da60f6a58de148b1dc72b37",
      "size": 856
    },
    "Identity": {
      "Build": [
        {
          "Ref": "u2qxcaw1xgwboenyvvwr1hsz6",
          "CreatedAt": "2026-04-17T17:03:05.363836323-05:00"
        }
      ]
    }
  }
]
juan@juan-virtualbox:~/Desktop/docker-lab/dockerfile-test$ |

```